Historical Relationships (2010-2017) Between Daily Changes in Yield Rates of US Treasuries and the 10-Year Bond

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1 Introduction

1.1 Motivation

Assume a portfolio of two financial assets X and Y, each of finite variance. If a proportion (or weight) $\alpha \in [0,1]$ of our portfolio's value is held in asset X with variance σ_X^2 and the remaining $1-\alpha$ in asset Y with variance σ_Y^2, then given a correlation between X and Y of ρ_{XY}, the variance of this portfolio is

$$\begin{aligned} Var[\alpha X + (1-\alpha)Y] &= Var[\alpha X] + Var[(1-\alpha)Y] + 2Corr[\alpha X, (1-\alpha)Y] \\ &= \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\rho_{XY}\sigma_X\sigma_Y \\ &= c_2\alpha^2 + c_1\alpha + c_0 \end{aligned}$$

where $c_2 := \sigma_X^2 + \sigma_Y^2 - 2\rho_{XY}\sigma_X\sigma_Y$, $c_1 := 2(\rho_{XY}\sigma_X\sigma_Y)(1-\sigma_Y)$, and $c_0 := \sigma_Y^2$ are known constants.

Since the last form shows a quadratic function of the parameter α, the optimal hedge ratio (which minimizes portfolio variance) is found either at the parabola's vertex (where the derivative w.r.t. α vanishes) or one of the endpoints (0 or 1). It is a common optimization problem in calculus.

Beyond the two-asset case above (for which we can apply $\hat{\rho}_{XY} = \hat{\beta}_1 \frac{\hat{\sigma}_X}{\hat{\sigma}_Y}$ using $\hat{\beta}_1$ from simple linear regression of Y upon X), bonds are used for hedging in portfolios with multiple (in some cases, well over 100) assets.

Suppose in another scenario on a fixed day, where we have a US Treasuries portfolio of $n-1$ assets each with weight α_i (which could be negative or greater than 1 if we had access to shorting and leverage in the futures market) and change in value X_i, we want to invest in an n-th asset (in this case, the 10-year bond) such that i) the expected value of change in the portfolio's total value is some constant α_0, and ii) the sum of squared residuals in our estimate for the n-th asset's change in value is minimized. The change in the portfolio's value due to asset i is $\alpha_i X_i$, hence we fix the expected value of the total change:

$$E[\sum_{i=1}^{n} \alpha_i X_i] = \alpha_0 \tag{1}$$

$$\sum_{i=1}^{n} \alpha_i \mu_{X_i} = \alpha_0 \tag{2}$$

$$\alpha_n \mu_{X_n} = \alpha_0 - \sum_{i=1}^{n-1} \alpha_i \mu_{X_i} \tag{3}$$

$$\mu_Y = \beta_0 + \sum_{i=1}^{n-1} \beta_i \mu_{X_i} \tag{4}$$

$$Y = \beta_0 + \sum_{i=1}^{n-1} \beta_i X_i + \varepsilon \tag{5}$$

where from line (3) to (4), we let $Y = \alpha_n X_n$, $\beta_0 = \alpha_0$, and $\beta_i = -\alpha_i$ (for $1 \le i \le n-1$); the error ε is a random variable with mean $E[\varepsilon] = 0$.

In particular, if $\beta_0 = 0$, then the expected value of change in the portfolio's value is zero. And more generally, this calls for a study of the relationship between the daily changes in yield rates (which are related to the prices) of US Treasuries within a linear regression context. If such a relationship exists, depending on the assumptions made, we may be able to predict and make inferences on the coefficients (from historical data).

1.2 Design

To clarify, the objective in this project is *not* to build an autoregressive model for forecasting time series into the the future, but rather to determine past structural relationships within fixed time cross-sections. I want to be precise because much interest from the literature in academia and the financial industry addresses the task of (after computing first differences on raw variables, e.g. price, interest rates, unemployment) modelling an unlagged response upon lagged regressors.

In this study as with almost any other on financial time series, while temporal effects will always violate i.i.d. assumptions, the actual degree of severity is the more practically relevant question; so by clever data partitioning, as well as examining correlation of partial residuals, we can allay at least some of our concerns.

My overall plan is to preprocess (treat, munge, clean) our data, analyze it from an exploratory perspective (for univariate and multivariate relationships), fit two regression models (after feature selection), and then compare their performance by out-of-sample evaluation.

1.3 Data Source

The raw data was downloaded from WRDS' dataset "Interest Rates (Federal Reserve, H15 report)", and which can be found at `https://wrds-web.wharton.upenn.edu/wrds/ds/frb/rates/index.cfm`

1.4 Summary of Data Files

By its order in the Preprocessing section, the following comma-separated value files are stored in the `data/` directory (all URL, file, and directory links are clickable in the electronic version of this document):

1. `daily_rates.csv` [input - original, raw data with missing observations, downloaded from WRDS]
2. `cleaned.csv` [intermediate - filtered out unwanted rows and columns from raw data]
3. `diffs.csv` [output - response and predictors extracted by arithmetic differencing], or alternatively `diffs_nodates.csv` [same differenced data, with first column (dates) removed]
4. `train.csv` and `test.csv` [response and predictors partitioned into two subsets], or alternatively `train_nodates.csv` and `test_nodates.csv` [ditto on partition, with first column (dates) removed]

1.5 Miscellaneous Files

These text files (besides `report.pdf`, which is located in the base directory) can be found in `misc/`:

- `misc/correlations.csv` (matrix of pairwise Pearson's ρ)
- `correlations.xlsx` (formatted matrices for Pearson's ρ, Kendall's τ, and their difference)
- `RMSE2.txt` (list of bootstrapped RMSEs for model with $d = 2$ predictors)
- `RMSE31.txt` (list of bootstrapped RMSEs for model with $d = 31$ predictors)
- `report.pdf` (this document)
- `report.tex` (LaTeX markup for generating this document)

Additionally, all graphics used to generate this report are saved in `misc/assets/`.

1.6 Programming Environment

I used R version 3.3.2 due to being familiar with its functionality:

Package	Functions	Explanation
leaps	regsubsets	all subsets and stepwise feature selection
lmtest	bgtest	Breusch-Godfrey's LM test
MASS	boxcox	Box-Cox power transformations
moments	kurtosis, skewness	kurtosis and skewness
multitaper	spec.mtm	spectral density (multitaper method)
pcaPP	cor.fk	correlation - fast Kendall's τ in O(nlogn)
stats	ks.test, qqline, qqnorm	Kolmogorov-Smirnov, quantile-quantile plotting
xts	diff.xts	differencing time series

1.7 Script Files

Included in `scripts/` directory for reproducibility and ordered by section number (not applicable to section 7):

1. `install.R`
2. `preprocess.R`
3. `exploratory.R`
4. `subsets.R`
5. `stepwise.R`
6. `evaluation.R`

2 Preprocessing

2.1 Handling of Missing Data

A preliminary treatment was conducted based on subject matter knowledge:

1. Filtered out all data prior to January 2010 due to structural instability from regime shift caused by regulatory changes (Dodd-Frank Act) following the financial crisis
2. Filtered out raw variables with more than 20% missing observations (or equivalently, less than 80% available over all business days)
3. Filtered out the raw variable DWPC_NA (Discount Window Primary Credit) because it is constant (exactly zero sample variance). We can begrudgingly work with low variance, but zero variance is an intractable problem technically because it leads to division by zero in matrix inversion, and substantially because the variable offers no predictive power.

2.2 Variable Naming Conventions

Here is a list of the selected variables organized by their prefixes (and what the Federal Reserve means by these acronyms), listed in alphabetical order:

- FCP_ = Financial Commercial Paper
- FF_ = Federal Funds (effective)
- LTAVG_ = Long-Term Average, inflation-indexed

- NFCP_ = Non-Financial Commercial Paper
- PRIME_ = bank PRIME loan
- TB_ = Treasury Bill
- TCMII_ = Treasury Constant Maturity, Inflation-Indexed
- TCMNOM_ = Treasury Constant Maturity, NOMinal

The suffixes 'WK', 'M', 'Y', 'P', 'NA' stand for 'week', 'month', 'year', 'plus', and 'not applicable' respectively. For instance, the variable FCP_M1 is the yield rate on Financial Commercial Paper of maturity 1 month. Similarly, LTAVG_Y10P is the average yield of bonds of maturity with 10 years or more.

2.3 Munging

After removing every invalid row with (at least one missing cell) from `daily_rates.csv`, I saved the (intermediate) result into `cleaned.csv`. The size of the new matrix is:

$$\text{number of observations, } n = 1340$$
$$\text{predictors, } d = 29$$

Then from `cleaned.csv`, I computed daily (arithmetic) differences (in yield rate) for each of 29 raw variables to generate 29 predictors. If we let the yield rate on day $t-1$ be r_t, then arithmetic difference is computed as

$$a_t = r_t - r_{t-1} \text{ (NA in the first row, for which } t = 1)$$

We can't use the logarithmic difference

$$l_t = \frac{r_t}{r_{t-1}} \text{ (better referred to as a ratio)}$$

due to presence of zero nominal and inflation-adjusted yield rates ever since the institution of quantitative easing. After removing the first row (whence the difference cannot be defined), the differences are stored in `diffs.csv`. So the size of the remaining matrix remains virtually unchanged:

$$\text{number of observations, } n = 1339$$
$$\text{predictors, } d = 29$$

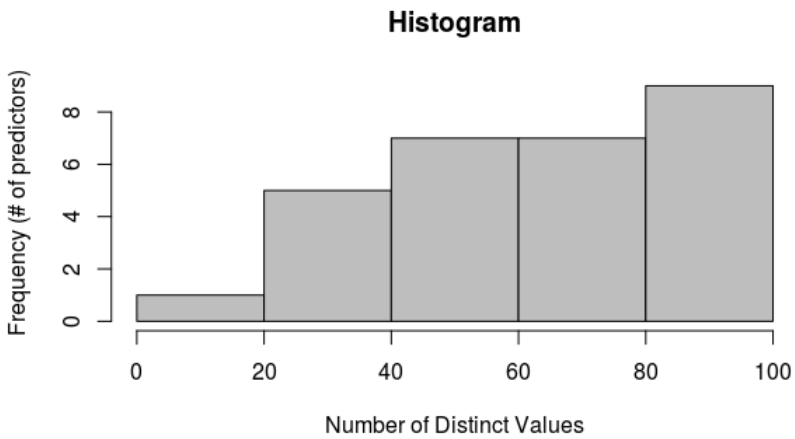
And let's check our rule of thumb for having sufficient observations, that is, the ratio of the number of observations to the number of regressors (predictors plus the intercept term):

$$\frac{n}{d+1} = \frac{1339}{29+1} \approx 45 \text{ observations per regressor}$$

This might better belong to EDA, but let's quickly check the number of distinct values per predictor (e.g. a sample with values of the sequence $\{-5, 0, 0, 5\}$ is said to have 3 distinct values):

```
    FCP_M1     FCP_M2     FCP_M3       FF_O LTAVG_Y10P    NFCP_M1    NFCP_M2    NFCP_M3   PRIME_NA      TB_M3      TB_M6
        63         61         72         27         82         45         47         44          2         34         35
    TB_WK4      TB_Y1  TCMII_Y10  TCMII_Y20  TCMII_Y30   TCMII_Y5   TCMII_Y7  TCMNOM_M1  TCMNOM_M3  TCMNOM_M6 TCMNOM_Y10
        48         44         78         85         82         89         84         46         35         36         86
TCMNOM_Y1 TCMNOM_Y20  TCMNOM_Y2 TCMNOM_Y30  TCMNOM_Y3  TCMNOM_Y5  TCMNOM_Y7
        45         95         63         80         73         97         91
```

Alternatively, a histogram (of the same information) for better visualization:



We see that all predictors have more than one distinct value (in fact, other than PRIME_NA, their sets of values are reasonably large - with cardinality of at least 27).

2.4 Justifications for Choice of Response

I believe that the change in nominal yield rate of the 10-year bond (TCMNOM_Y10) is a good dependent variable for modelling, and my (practical, informal) reasons are twofold:

1. First, it has seen high year-to-date trading volume on the CME (Chicago Mercantile Exchange's) short-list of popular futures and options at `http://www.cmegroup.com/daily_bulletin/monthly_volume/Web_Volume_Report_CMEG.pdf`. The 10-year bond's representative security has more participants in the markets, in which prices (affecting yield rates) tend to be more efficient so that errors will be lower than for underlying instruments with very illiquid (or non-publicly traded) assets.

2. As a matter of fact, there is more general interest in analyzing bonds with 10 to 30-year maturities. For rough reference, here are search result counts on Google Scholar for "[Maturity] bonds" (e.g. "10 year bonds")

[Maturity]	Number of search results
1 month	12
2 month	4
3 month	90
6 month	44
1 year	275
2 year	312
3 year	276
5 year	865
7 year	169
10 year	3460
20 year	1010
30 year	1890

The 2-year bond had the least interest, while the highest belonged to the 10-year.

2.5 Autocorrelation Test (#1)

In our original equation

$$Y_t = \beta_0 + \sum_{i=1}^{d} \beta_i X_{t,i} + \varepsilon_t \ (t = 1, \ldots, n)$$

an AR(L) model

$$\varepsilon_t = \gamma_0 + \sum_{j=1}^{L} \gamma_i \varepsilon_{t-j} + \upsilon_t$$

can be fit on the residuals ε_t so we can perform another ordinary least-squares regression

$$\varepsilon_t = \gamma_0 + \sum_{k=1}^{L} \gamma_k \varepsilon_{t-k} + \sum_{k=L+1}^{L+d} \gamma_k X_{t,k} + \upsilon_t$$

and F-test the fitted autoregressive coefficients $\gamma_1, \ldots, \gamma_L$ between the AR(L) and the combined equation; this idea owes credit as Breusch-Godfrey's LM.

On our data in `diffs.csv`, I obtained F-statistic and p-value

$$F_{5,1305} = 13.95$$
$$p < 10^{-12}$$

therefore, the null hypothesis, $H_0 : \gamma_1 = \cdots = \gamma_L = 0$, can be rejected at any sane level – with virtually no chance of Type I error, we can conclude that there is definitely autocorrelation.

2.6 Partitioning

In order to deal with autocorrelation and guard against overconfidence from overfitting, let's partition the data into training and test subsets (saved as `train.csv` and `test.csv`). We can assign each odd-numbered observation into the training subset, and each even-numbered observation into the test subset. Number-theoretically, this is a non-random procedure; but the number of subpopulations is 2, which is coprime with 5 (business) or 7 days in a week (sidestepping the possibility of having certain days of a week, e.g. all Mondays, allocated to, say, the training subset); any systematic pattern would be further noised by holidays.

To be extra safe, we can perform a two-sample, two-sided Kolmogorov-Smirnov test of identical distributions on each corresponding pair of predictors between the training and test subsets. For empirical distribution functions $\hat{F}_1$ and $\hat{F}_2$ with m and n data points respectively, the KS statistic denoted D is the supremum of absolute difference in cumulative proportions

$$D = \sup_{x \in \mathbb{R}} |\hat{F}_1(x) - \hat{F}_2(x)|$$

which asymptotically satisfies the property

$$Pr[D > \sqrt{-\tfrac{m+n}{2mn} ln \tfrac{\alpha}{2}}] = \alpha$$

for significance level α. In our case, the training and test subsets have essentially the same number of observations ($m = 670 \approx 669 = n$), which means that after rearranging, the p-value can be extracted

$$p = 2e^{-nD^2}$$

and exhibited below for each variable:

```
    FCP_M1     FCP_M2     FCP_M3       FF_O LTAVG_Y10P    NFCP_M1    NFCP_M2    NFCP_M3   PRIME_NA      TB_M3      TB_M6
0.25209782 0.16767549 0.30670529 0.99999997 0.97185248 0.05060711 0.17036955 0.52755725 1.00000000 0.99947127 0.09971118
    TB_WK4      TB_Y1  TCMII_Y10  TCMII_Y20  TCMII_Y30   TCMII_Y5   TCMII_Y7  TCMNOM_M1  TCMNOM_M3  TCMNOM_M6 TCMNOM_Y10
0.17630297 0.13168673 0.93113351 0.61803034 0.99996619 0.99882950 0.99894979 0.15616893 0.99897037 0.07584409 0.47747741
 TCMNOM_Y1 TCMNOM_Y20  TCMNOM_Y2 TCMNOM_Y30  TCMNOM_Y3  TCMNOM_Y5  TCMNOM_Y7
0.21283776 0.70040548 0.96630345 0.97853484 0.58152689 0.70333900 0.72457682
```

none of which suggest rejecting the null hypothesis at an $\alpha = .05$ level (not even discounting for family-wise error rate).

2.7 Autocorrelation Test (#2)

Now that we have partitioned the data, let's perform Breusch-Godfrey's LM test on each subset

$$\text{training subset: } F_{5,636} = 0.35, p = 0.88$$
$$\text{test subset: } F_{5,635} = 1.77, p = 0.12$$

This time, the evidence against the null hypothesis is weak. Hence, while it is not strictly necessary to check prior to performing regression modelling, we have dealt away with autocorrelation in the response early on (lest we have to rewind later on), and will verify again on the model in a subsequent section of this paper.

3 Exploratory Analysis

Note: These procedures are conducted solely on the training subset to mitigate excessive data snooping.

3.1 Univariate

Let's examine the distributions of each predictor and the response to see if there's anything unusual (e.g. sparse outliers). I will mostly eschew histograms for these applications because the visual display relies heavily on the number of bins or breaks used; it's more straightforward to observe some sample statistics and reference the frequency tables when the number of values is reasonably bounded.

The skewnesses were computed to be:

```
     FCP_M1      FCP_M2      FCP_M3        FF_O  LTAVG_Y10P     NFCP_M1     NFCP_M2     NFCP_M3    PRIME_NA       TB_M3
 0.28816460  0.45509787  0.30202314  4.95578516 -0.55472655 -0.98210883 -0.24103789 -0.07526206 25.82637208  0.42928801
      TB_M6      TB_WK4       TB_Y1   TCMII_Y10   TCMII_Y20   TCMII_Y30    TCMII_Y5    TCMII_Y7   TCMNOM_M1   TCMNOM_M3
-0.98125110  3.42858628  0.58161976 -0.14833162 -0.22201717 -0.60874354 -0.45430138 -0.49287860  3.78016335  0.55834482
  TCMNOM_M6  TCMNOM_Y10   TCMNOM_Y1  TCMNOM_Y20   TCMNOM_Y2  TCMNOM_Y30   TCMNOM_Y3   TCMNOM_Y5   TCMNOM_Y7
-0.92274355 -0.36429756 -0.12509411 -0.49613963 -0.46088561 -0.54795145 -0.34630539 -0.29287081 -0.29443609
```

Most of the variables are midly skewed to the left ($\frac{M_3}{\sigma^3} < 0$, where M_3 is the third central moment). A few anomalies we will look into shortly are PRIME_NA (25.8), FF_O (5.0), and TB_WK4 (3.4).

Proceedingly, the excess kurtoses were computed:

```
   FCP_M1     FCP_M2     FCP_M3       FF_O LTAVG_Y10P    NFCP_M1    NFCP_M2    NFCP_M3   PRIME_NA      TB_M3      TB_M6
 4.808091   4.449388   4.670653  99.757353   4.346156   8.990638   2.195612   1.917406 665.001495   7.091860  21.093541
   TB_WK4      TB_Y1  TCMII_Y10  TCMII_Y20  TCMII_Y30   TCMII_Y5   TCMII_Y7  TCMNOM_M1  TCMNOM_M3  TCMNOM_M6 TCMNOM_Y10
39.015071   9.877976   3.173159   3.346411   4.785723   4.862440   3.698230  45.661439   8.052380  19.986117   4.192839
TCMNOM_Y1 TCMNOM_Y20  TCMNOM_Y2 TCMNOM_Y30  TCMNOM_Y3  TCMNOM_Y5  TCMNOM_Y7
 8.924106   4.954702   2.624553   5.613771   2.751703   3.000730   3.632767
```

All of these are leptokurtic ($\frac{M_4}{\sigma^4} - 3 > 0$, where M_4 is the fourth central moment) with fatter tails than normal. In particular, PRIME_NA (665.0), FF_O (99.8), TCMNOM_M1 (45.7), TB_WK4 (39.0), and TB_M6 (21.1) have very high excess kurtoses of greater than 20.

Among the values observed for PRIME_NA, there are 669 0's and one 0.25. There is unlikely to be any significance for such an invariant dummy/indicator predictor. Any reasonable feature selection algorithm should be able to remove it anyway, for not reducing SSE by much.

Here is a table of the values (with frequencies below) for FF_O:

```
-0.1 -0.09 -0.08 -0.07 -0.06 -0.03 -0.02 -0.01     0  0.01  0.02  0.03  0.05  0.06  0.09  0.25
   1     2     2     1     1     2    13    94   451    77    12     7     2     3     1     1
```

The distribution has a large peak at the origin (which means that the yield rate remains unchanged over most days), its density quickly falls off on both sides, and stretches out to long tails. The separation between consecutive values is at most 0.03, except at the right tail where there is a jump between the second-highest value of 0.09 and the highest of 0.25. These highest right-tailed outlier occured is the 586th observations (in the training subset), corresponding to December 15, 2016.

For TB_WK4, we see a somewhat similar distribution:

```
-0.13 -0.11 -0.09 -0.08 -0.07 -0.06 -0.05 -0.04 -0.03 -0.02 -0.01     0  0.01  0.02  0.03  0.04  0.05  0.06  0.07  0.08
    1     1     1     1     4     9     7     7    20    45   135   246   109    28    26     6     5     5     3     1
 0.09   0.1  0.12  0.14  0.19  0.34
    5     1     1     1     1     1
```

Once again, the second-highest value here of 0.19 is separated from the highest of 0.34. The highest observations was numbered 645th, with date September 5, 2017.

And we see a similar distribution for TCMNOM_M1:

```
-0.13 -0.11 -0.08 -0.07 -0.06 -0.05 -0.04 -0.03 -0.02 -0.01     0  0.01  0.02  0.03  0.04  0.05  0.06  0.07  0.08  0.09
    1     1     2     2     8     6    10    18    41   132   259   103    38    18     9     8     2     5     1     2
 0.13  0.14  0.19  0.34
    1     1     1     1
```

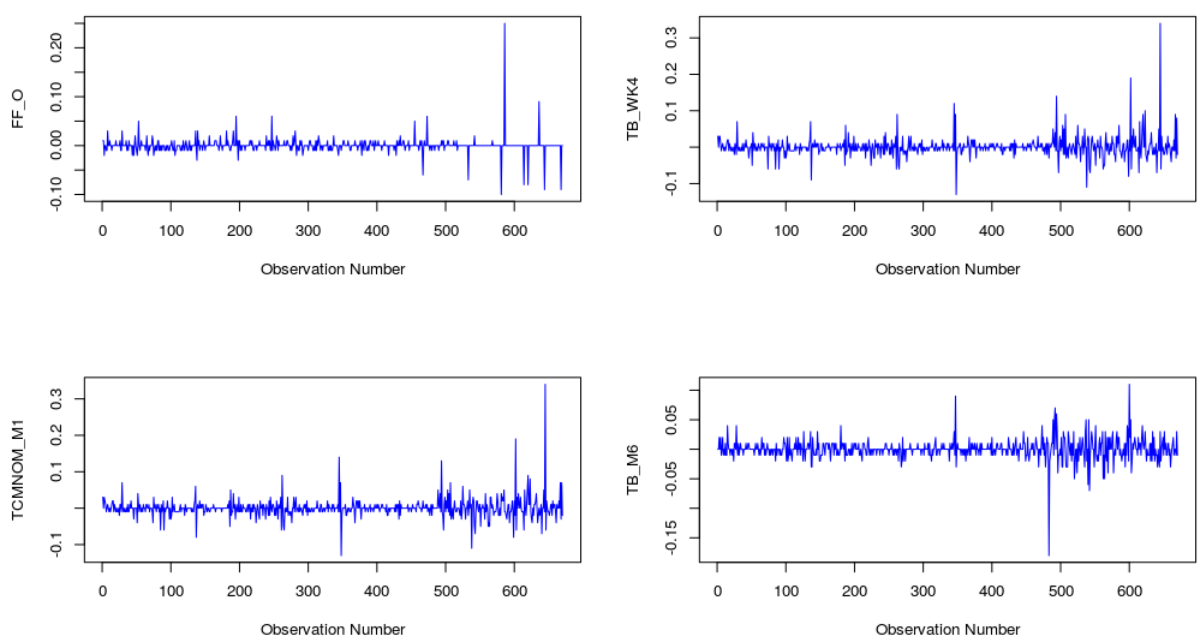
This time, it is again the 645th observation with the highest single-day value (largest increase in yield rate), on September 5, 2017.

For TB_M6:

```
-0.18 -0.07 -0.06 -0.05 -0.04 -0.03 -0.02 -0.01     0  0.01  0.02  0.03  0.04  0.05  0.06  0.07  0.09  0.11
    1     1     1     3     5    18    38   140   266   117    43    21     7     5     1     1     1     1
```

The most extreme value is on the lower tail at -0.18 (which is -0.11 lower than the next most negative value). This is the 483rd observation on September 23, 2015.

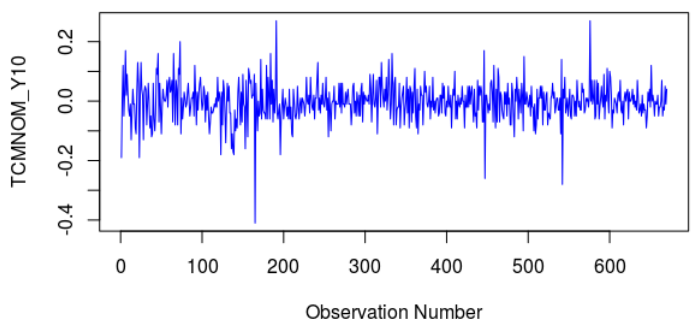
By plotting these four predictors' time series, we notice two phenomena:



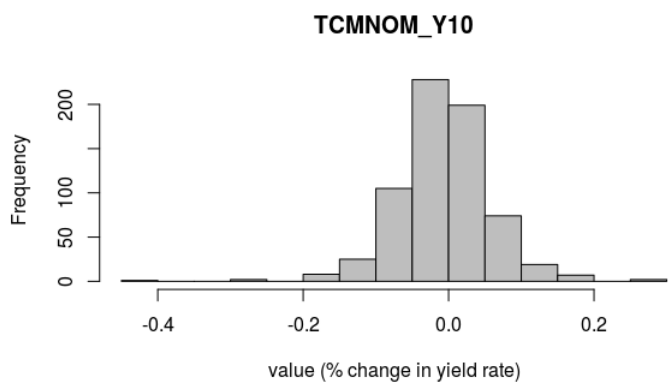
First, the most extreme value in each case is indeed much greater in magnitude than most of the other values. Second, they tend to occur around other spikes (in the manner of earthquakes measured on a seismograph); this has been referred to as "volatility clustering" by Mandelbrot. While it would be tenuous to seek evidence for causality with our observational data, some possible explanations for these high volatility events may include the EU's sanctioning Ukraine amid the War in Donbass on December 15, 2016; North Korea's hydrogen bomb testing and Trump's announcement of ending DACA on September 5, 2017; and the EU

forcing member nations to take in migrants from Greece and Italy on September 23, 2015. Rigorous causal inference would stretch beyond the scope of this study. I'd argue against immediately removing outliers from regression because there are likely to be real economic explanations. But we can keep this thought in the back of our mind in light of standard errors when performing OLSR.

In any case, let's look at the response, TCMNOM_Y10, by plotting the time series



and histogram of observed values:



While it is slightly negatively skewed and leptokurtic, at least its distribution is relatively closer to normal than most of the predictors we just scrutinized. This is only a cursory mention – we will return to the topic of transformations near the end of this section.

3.2 Bivariate

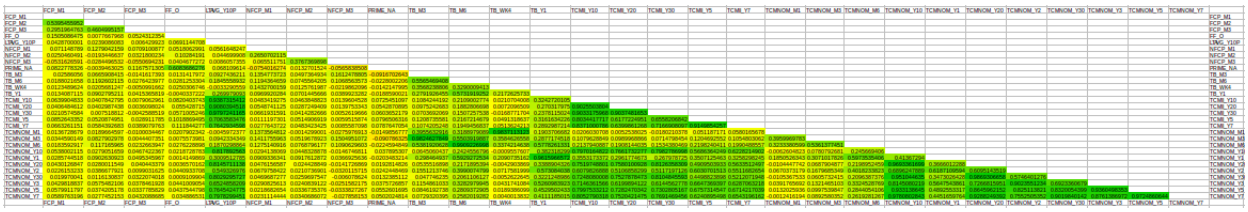
I computed the sample correlations (Pearson's ρ) between each pair u and v (to avoid confusion with random variables X and Y, these are any two columns in matrix of observations)

$$\rho = \frac{1}{(n-1)s_u s_v}[(\sum_{i=1}^{n} u_i v_i) - n\bar{u}\bar{v}]$$

and saved an unformatted matrix as `correlations.csv`.

Since correlation is symmetric with respect to its arguments, the matrix is symmetric; I removed the upper-right triangle of the matrix to reduce clutter. Also, there is no benefit to keeping the diagonals since they are always unitary. The labels were duplicated on the bottom- and right-hand sides to save the reader from constant scrolling. The formatted result is saved in Excel spreadsheet format as a copy named `correlations.xlsx` (the first sheet, `Pearson's` ρ) for convenience of highlighting and visualization.

Here is an extremely down-scaled version (bird's eye view) of the matrix for Pearson's ρ:



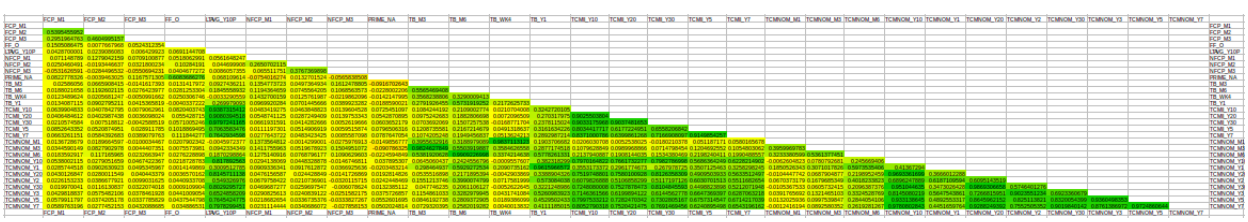
From the color scale (red at -1, to yellow at 0, to green at 1), we see few negative correlations. Most of the cells are yellow, and I see a few patches of green. Near the left area, the predictor LTAVG_Y10P is positively correlated with most of the TCMII_ and TCMNOM_ of longer maturities (Y5 and above). This makes sense intuitively because LTAVG_Y10P is a weighted average of bonds of maturities at least 10 years. I would not be surprised if it ends up being removed by feature selection. In the middle and right areas, the other clusters of positive correlations are between TCMII_ and TCMNOM_, i.e. pairs among TCMII_ and TCMII_; TCMNOM_ and TCMNOM_; TCMII_ and TCMNOM_. The latter can be rationalized because changes in inflation rates (the difference between TCMII_ and TCMNOM_) are generally lower than the changes in both real and nominal yield rates. The aforementioned areas of intersection are mostly but not entirely green though, as bonds of different maturities tend to have lower correlation (that is, they are closer to being yellow or zero, having lower correlations).

These correlations confirm our subject matter knowledge regarding the underlying financial assets. Meanwhile, investors, hedgers, and speculators have preferences toward certain maturities, so if mispricings become very large, they will seize the opportunity on undervalued and overvalued securities. Along with arbitrageurs (including but not limited to marketmakers), these competitive forces push the market toward equilibrium, at least until no more low-risk profit can be extracted (after discounting for fees). In view of the non-spuriously high correlations, we should not hesitate to remove predictors from the model if our feature selection algorithm recommends it.

The rank-based statistic Kendall's τ can measure non-linear correlations, and is defined by

$$\tau = \binom{n}{2}^{-1}(n_= - n_\neq)$$

where cardinalities $n_= = |\{(u_i, v_i) \mid (u_i - v_i)(u_j - v_j) > 0\}|$ and $n_\neq = |\{(u_i, v_i) \mid (u_i - v_i)(u_j - v_j) < 0\}|$ are the number of concordant and discordant pairs, respectively. I computed and saved these metrics (second sheet in `correlations.xlsx`, titled `Kendall's` τ) as well:



Upon a cursory glance, they appear similar. A difference matrix is saved in the third sheet in the same file, where we can see that all differences in correlation have magnitudes less than 0.30, save for one exception – between $u = PRIME_NA$ and $v = FF_O$, Pearson's ρ is much higher than Kendall's τ and this can be understood in terms of their formulae.

In our formula for Pearson's ρ, the summand

$$\sum_{i=1}^{n} u_i v_i = u_{586} v_{586} + \sum_{i \neq 586} u_i v_i = (0.25) v_{586} + \sum_{i \neq 586} (0) v_i = (0.25) v_{586}$$

is dominated by the 586th observation. The values of $\bar{u}$ and $\bar{v}$ are small, so the multiplicand $(\sum_{i=1}^{n} u_i v_i) - n\bar{u}\bar{v}$ is large and positive. Likewise, the sample standard deviations s_u and s_v are small, so the denominator of the multiplier $(n-1)s_u s_v$ is small, whereby the multiplier is large. Consequently, Pearson's ρ is largely positive (0.61).

On the other hand, Kendall's τ has mostly ties (counted as neither concordant nor discordant pairs) between the $\binom{670}{2} = 224{,}115$ values of 0 for PRIME_NA makes the multiplicand $n_= - n_{\neq}$ close to 0 (consequently, $\tau = 0.05$ to be more precise). In such a sense, Kendall's τ can be said to be more robust to outliers than Pearson's ρ.

As for our response, TCMNOM_Y10 is positively correlated with most of the longer maturity TCMNOM_ and TCMII_ predictors, along with LTAVG_Y10P due to fundamental economic relationships. Its correlations with bonds of lower maturities is much lower, and in fact, close to zero.

3.3 Multivariate

It is already evident that without feature selection, we will suffer from a multicollinearity problem. For a quantitative evidence within a multiple regression framework, we can compute the variance inflation factors, defined by

$$VIF_i = (1 - R_i^2)^{-1}$$

where R_i^2 for each predictor X_i is the proportion of variance explained in the equation

$$X_i = \alpha_0 + \sum_{j \neq i} \alpha_j X_j + \varepsilon$$

The VIFs computed were

FCP_M1	FCP_M2	FCP_M3	FF_O	LTAVG_Y10P	NFCP_M1	NFCP_M2	NFCP_M3	PRIME_NA	TB_M3	TB_M6
1.532622	1.767220	1.339126	1.711048	43.752457	1.191589	1.328644	1.252232	1.700126	32.154905	61.220657
TB_WK4	TB_Y1	TCMII_Y10	TCMII_Y20	TCMII_Y30	TCMII_Y5	TCMII_Y7	TCMNOM_M1	TCMNOM_M3	TCMNOM_M6	TCMNOM_Y1
36.384572	13.873710	21.809883	9.359842	33.418765	8.497559	9.472820	37.064921	31.241210	60.436510	15.462439
TCMNOM_Y20	TCMNOM_Y2	TCMNOM_Y30	TCMNOM_Y3	TCMNOM_Y5	TCMNOM_Y7					
67.528869	6.514303	50.307362	16.504496	47.129205	66.453660					

where we can see that the predictors with highest VIFs are prefixed by TCMNOM_, TCMII_, and TB_. They are well above the rule of thumb whereby VIF > 10 suggests high multicollinearity.

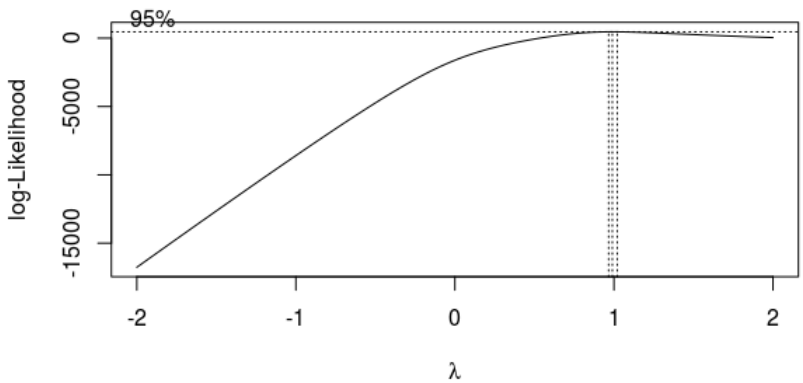
3.4 Response Transformation

Should we apply variance stabilization? Let's consider a power transform on $v \in \mathbb{R}_+^n$ with parameter λ

$$\psi_\lambda(_i) = \begin{cases} \frac{v_i^\lambda - 1}{\lambda(\tilde{v})^{\lambda-1}}, & \text{for } \lambda \neq 0 \\ \tilde{v} ln(v_i), & \text{if } \lambda = 0 \end{cases}$$

employing the notation $\tilde{v} = (\prod_{i=1}^{n} v_i)^{1/n}$ for the geometric mean. Now, our original response y contains negative values, so we shift it, $v_i = y_i + \delta$, by $\delta = \max_{1 \leq i \leq n} (-y_i) = 0.41$ (this more general form is referred to as the Box-Cox transformation).

We see in the plot of log-likelihood for our single parameter

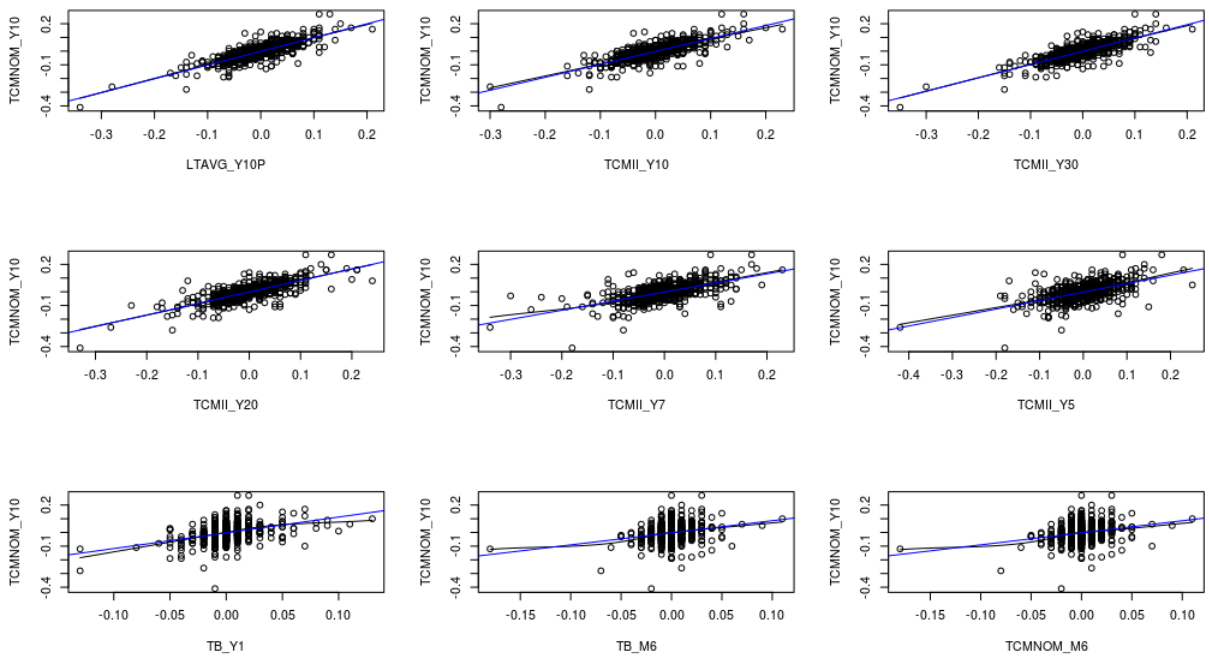


the 95% confidence interval (dotted) of the MLE contains the unit, $\lambda_0 = 1$, ergo it is better to leave the response untransformed.

3.5 Predictors Transformation

Should we transform the predictors? This should really be performed by examining partial residual plots (in the context of multiple regression), although my intuition thinks that it may be conducive to start with model specification earlier. If we don't transform the predictors, then we are implying that the relationships are linear (which is an assumption in itself, although admittedly it is usually a good one).

In decreasing order of Kendall's τ, the predictors most correlated to TCMNOM_Y10 and their correlation coefficients are: LTAVG_Y10P (0.64), TCMII_Y10 (0.62), TCMII_Y30 (0.61), TCMII_Y20 (0.59), TCMII_Y7 (0.46), TCMII_Y5 (0.40), TB_Y1 (0.26), TB_M6 (0.15), and TCMNOM_M6 (0.15). Of course, there is no guarantee that these will be the most important predictors in multiple regression; but at this stage, let's work with what we have and plot them against the response (each with black loess smoother at 2/3 span overlayed in blue by a line of best fit):



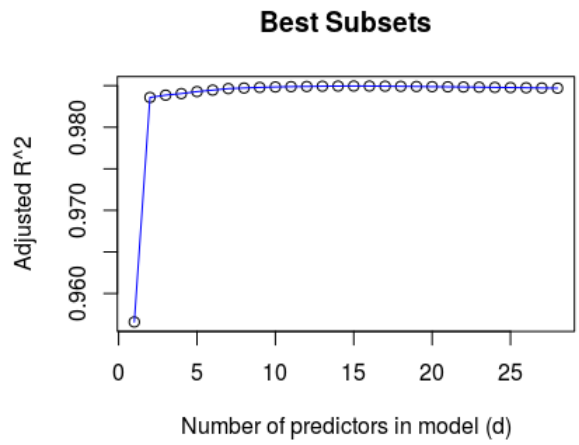
By this naive approach of model pre-specification, one can tell from a glance that linearity is reasonably good in most cases (the loess smoothers are not far off from the lines of best fit). I am slightly tempted to apply quadratic transformations on TCMII_Y7 and TB_Y1, but it seems overkill and especially risky considering the leptokurtic distributions of these predictors; the presence of outliers may lead to unstable fits with more severely biased coefficients, albeit there isn't serious cause for removing the outliers either.

4 All Subsets Feature Selection

Note: These procedures are conducted solely on the training subset to mitigate excessive data snooping.

4.1 Outcome

For n observations and d features, the worst-case time complexity for all subsets is asymptotically upper-bounded by $O((2^d)nd^2)$. Since we have moderate n and d, all subsets model selection is appropriate. I found that the adjusted R^2 is maximized at 0.9850 with $d = 15$ predictors.



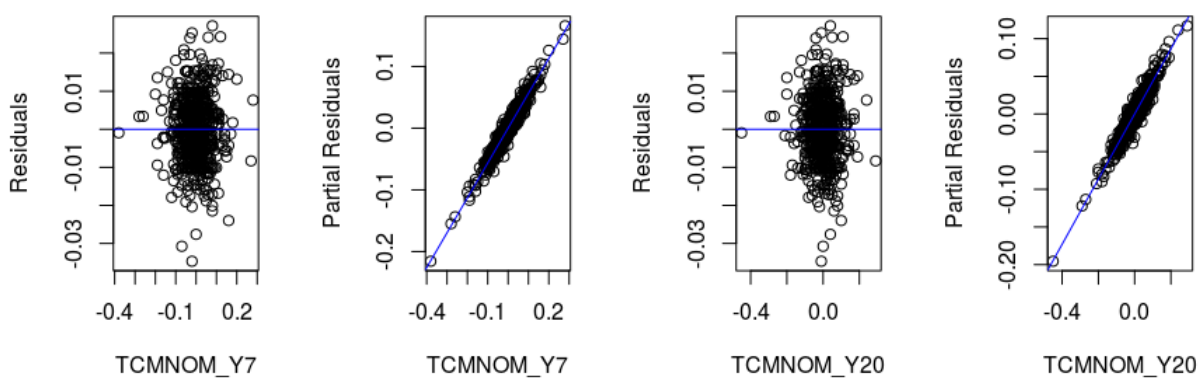
However, from the plot above of number of predictors used versus the best model's adjusted R^2, we can visually identify an approximate plateau at $d = 2$ with adjusted R^2 at 0.9836. This is not surprising considering the high correlation between our predictors. I personally prefer more parsimonious models for sake of interpretability.

4.2 Features

For $d = 2$ (two predictors plus an intercept term), the best model selected includes TCMNOM_Y7 and TCMNOM_Y20. So let's run OLSR using these predictors.

The Gauss-Markov theorem, of OLS giving the BLU estimators of coefficients, depends upon the errors (estimated by the residuals) satisfying three properties: linearity, independence, and homoscedasticity.

4.3 Gauss-Markov Assumptions - Linearity



From the residual versus predictors and partial residual plots, no nonlinearities can be identified.

4.4 Gauss-Markov Assumptions - Independence via Autocorrelation Test (#3) and MTM

First, let's call upon Breusch-Godfrey's LM test to check for autocorrelation (up to order 5):

$$F_{5,662} = 0.55$$
$$p = 0.74$$

There is insufficient evidence at any reasonable level to reject the null hypothesis (of zero autocorrelation).

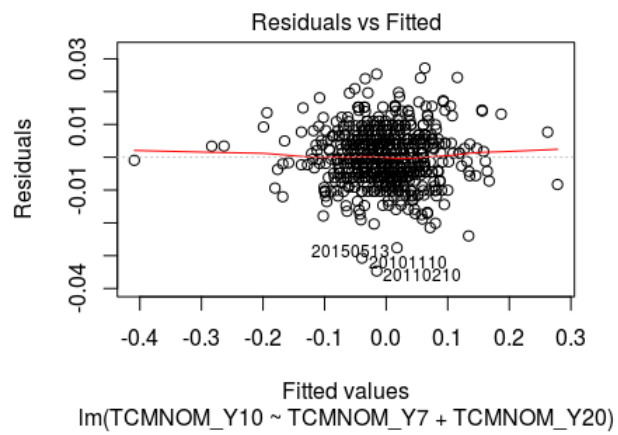
Second, let's apply the MTM (multitaper method) from spectral analysis and test the ratio of variance at specified frequencies versus white noise, since a periodic signals's Fourier transform approaches Dirac's delta function of large magnitude at its frequencies. I'll use $k = 30$ tapers for more power than the default suggested by the multitaper package in R, since we are dealing with a decent sample size over a moderately long time period

$$\text{for 5 days (one business week): } F_{2,58} = 0.70$$
$$\text{for 22 days (one business month): } F_{2,58} = 2.81$$

With $\nu_1 = 2$ and $\nu_2 = 2d - 2 = 58$ degrees of freedom, the critical value for F_{ν_1,ν_2} at $\alpha_2 = .025$ is 3.93 (I am using the Dunn–Šidák correction $\alpha_d = 1 - (1 - \alpha_1)^{1/d}$ with a a single test threshold of $\alpha_1 = .05$ and d=2 tests). That is to say, we cannot reject the null hypothesis of non-periodicity (otherwise if we found sufficient evidence, it would have justified a two-stage approach or detrending with a moving average – before regression, and then afterward, adding back the trend).

4.5 Gauss-Markov Assumptions - Homoscedasticity

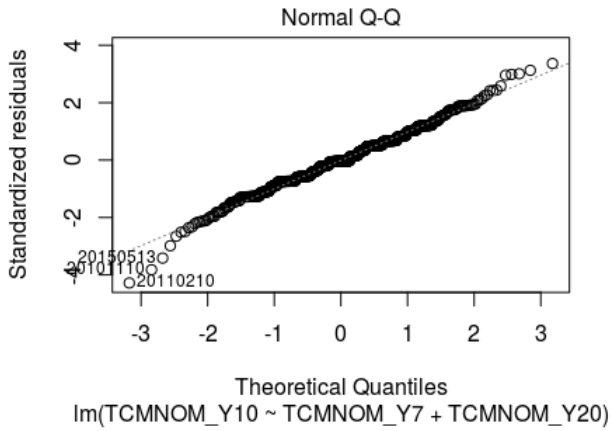
Plotting the residuals versus fitted values,



the residuals appear to be randomly distributed with roughly constant variance for fitted values in the middle section (between -0.1 and $+0.1$); outside this range, there is not a terrible amount of data to draw meaningful conclusions.

4.6 Normality of Residuals

Having verified Gauss-Markov assumptions, let's proceed by checking the normality of the residuals (so that inferences on the coefficients are valid)



According to the normal quantile-quantile plot, the distribution of ε seems close enough to normal.

4.7 Model

From the regression output, I found that both x_1 =TCMNOM_Y7 and x_2 =TCMNOM_Y20 are significant in predicting the response y =TCMNOM_Y10 with respective t-values of 42.4 and 33.2 respectively (their p-values are less than 10^{-15}). The intercept is not very significant, however, with $t = 0.99$ and $p = 0.322$. Our fitted equation is

$$\hat{y} = 0.0003105 + 0.5651607x_1 + 0.4324564x_2$$

The regression coefficients can be interpreted as such. If the 7-year and 20-year nominal yield rates remain unchanged (0% difference), then the 10-year nominal year rate is also expected to stay roughly the same (increase of 0.03%, albeit not significant). A 1% change in the 7-year nominal yield rate is expected to correspond with a 0.56% change in the 10-year nominal yield rate (holding the change in 20-year nominal yield rate constant). A 1% change in the 20-year nominal yield rate is expected to correspond to a 0.43% change in the 10-year nominal yield rate (holding the change in 7-year nominal yield rate constant).

From the standard errors of the coefficients, we can construct 95% confidence intervals of the coefficients; if the regression assumptions are valid, then over a large number of regressions conducted, 19 out of every 20 such confidence intervals will contain the actual coefficients of the true best fit. I present them in a table:

```
                    2.5 %       97.5 %
(Intercept) -0.0003049199 0.0009258891
TCMNOM_Y7    0.5389580926 0.5913633987
TCMNOM_Y20   0.4068509228 0.4580618292
```

While the 95% confidence interval for the intercept contains zero, both confidence intervals for the predictors are strictly positive (right of the origin). Despite the outliers we noted earlier, the standard errors are small relative to the (magnitudes of the) coefficients.

This model has an RMSE of 0.008089.

5 Stepwise Selection

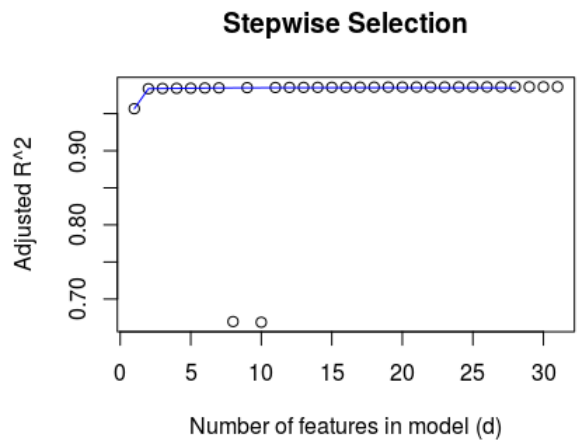
Note: These procedures are conducted solely on the training subset to mitigate excessive data snooping.

5.1 Methodology

A type of diagnostic procedure enabled by sequential replacement (bidirectional search – both forward and backward selection) involves incorporating 2nd-order polynomial and interaction terms (henceforth to be referred to as features). That is, we run forward selection with backward elimination of insignificant features of the form X_iX_j, where X_i and X_j are chosen from among our original predictors (they may be the same predictor, giving rise to a quadratic term).

5.2 Outcome

I ran the process and plotted the adjusted R^2 against the number of features allowed in the model:



The resulting graph looked like what we saw before with the untransformed predictors. Furthermore, the maximum adjusted R^2 found of 0.9862 with 31 features is only a slight improvement over the adjusted R^2 found in the previous section of 0.9836. I am not quite sure why there are two points below 0.70 for $d = 8$ and $d = 10$.

5.3 Features

Interestingly, for up to $d = 3$ features, the models selected were the same as the ones we saw with all subsets selection on only the original predictors in the previous section

Number of features	Features Included in Best Model
$d = 1$	TCMNOM_Y7
$d = 2$	TCMNOM_Y7, TCMNOM_Y20
$d = 3$	TCMNOM_Y7, TCMNOM_Y20, TCMNOM_Y30

Only with $d = 4$ and beyond, an interaction term, NFCP_M2 : TCMNOM_Y1, enters the equation (the colon ':' denoting Hadamard, or element-wise, multiplication). For these reasons, I would make the case that the decision to exclude non-linear terms offers an adequate model.

5.4 Model

However, for sake of comparison and in demonstration of concept, it might be worthwhile to compare a couple of our models out-of-sample in the subsequent section. For stepwise selection, we will use the model with lowest adjusted R^2. Since I am not trying to make any formal statistical inferences in this case, let's jump past the formalities (in the next section, I will employ a technique that is independent of parametric assumptions). Below is the output from console:

```
Call:
lm(formula = form31, data = train)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.031714 -0.004700  0.000002  0.004617  0.027114 

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)    
(Intercept)           0.0001853  0.0003222   0.575 0.565563    
LTAVG_Y10P            0.0699827  0.0371295   1.885 0.059908 .  
NFCP_M1              -0.0371192  0.0124472  -2.982 0.002972 ** 
NFCP_M2               0.0474743  0.0127305   3.729 0.000209 ***
TCMII_Y10             0.0795048  0.0210054   3.785 0.000168 ***
TCMII_Y30            -0.1114801  0.0310906  -3.586 0.000362 ***
TCMII_Y5             -0.0244220  0.0097402  -2.507 0.012412 *  
TCMNOM_Y20            0.2968338  0.0347657   8.538  < 2e-16 ***
TCMNOM_Y2            -0.0268022  0.0157419  -1.703 0.089129 .  
TCMNOM_Y30            0.1375901  0.0314392   4.376 1.41e-05 ***
TCMNOM_Y7             0.5629136  0.0191624  29.376  < 2e-16 ***
NFCP_M1:FCP_M2        0.4190454  0.2376127   1.764 0.078284 .  
FCP_M2:TB_M3          0.7041094  0.3446893   2.043 0.041489 *  
FCP_M3:NFCP_M3       -0.4585357  0.2046326  -2.241 0.025384 *  
TCMII_Y5:FCP_M3       0.8149317  0.3525686   2.311 0.021128 *  
FCP_M3:TCMII_Y7      -0.5889374  0.3412975  -1.726 0.084907 .  
TCMNOM_Y2:FCP_M3     -0.7255319  0.2409015  -3.012 0.002700 ** 
NFCP_M2:TB_Y1        -5.2987345  1.9741579  -2.684 0.007462 ** 
NFCP_M2:TCMNOM_Y1     7.0943908  2.0008106   3.546 0.000420 ***
NFCP_M3:TB_Y1         3.0710932  1.6626880   1.847 0.065201 .  
NFCP_M3:TCMNOM_Y1    -4.2031245  1.8447230  -2.278 0.023029 *  
TB_M3:TCMNOM_M3      -0.9248227  0.4985767  -1.855 0.064068 .  
TCMNOM_Y2:TB_M3       1.4208436  0.6769994   2.099 0.036233 *  
TB_M6:TCMNOM_M6       0.9262685  0.3938485   2.352 0.018984 *  
TB_M6:TCMNOM_Y5       1.6210707  0.7116397   2.278 0.023060 *  
TCMII_Y7:TB_WK4      -0.3957426  0.2520989  -1.570 0.116960    
TCMII_Y7:TB_Y1        2.1527934  0.4792296   4.492 8.37e-06 ***
TB_Y1:TCMNOM_M3      -1.6297047  0.7513404  -2.169 0.030447 *  
TCMNOM_M3:TCMII_Y20  -2.2898589  0.9114849  -2.512 0.012243 *  
TCMII_Y30:TCMNOM_M3   2.2868735  0.9874086   2.316 0.020872 *  
TCMII_Y5:TCMNOM_Y1   -2.2554005  0.5857129  -3.851 0.000130 ***
TCMNOM_Y20:TCMNOM_M6 -1.7795808  0.5997226  -2.967 0.003116 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.007446 on 638 degrees of freedom
Multiple R-squared:  0.9868,	Adjusted R-squared:  0.9862 
F-statistic:  1538 on 31 and 638 DF,  p-value: < 2.2e-16
```

In this model, the two most significant features are TCMNOM_Y7 (t=29.38) and TCMNOM_Y20 (t=8.54), which are the same two predictors used for the model in Section 4. Many of the other coefficients are non-significant at $\alpha = .01$, and it is not clear how to interpret such terms as NFCP_M2 : TCMNOM_Y1. The RMSE is 0.007266, which is lower than that of the d=2 model (chosen from a smaller family) in the previous section. But let's see how they perform out-of-sample, which is the real question.

6 Out-of-Sample Evaluation

Note: In order to ensure integrity by unbiased estimates of error, these procedures are conducted solely on the test subset.

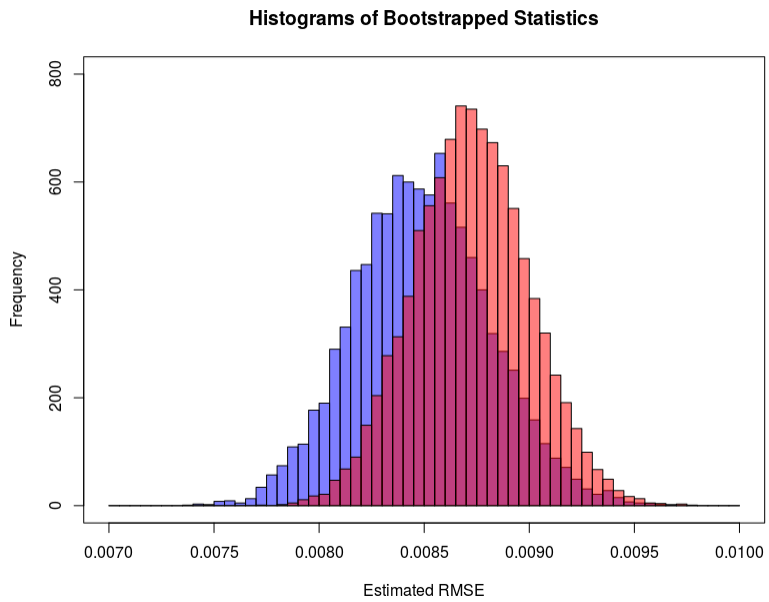
6.1 Methodology

I will fix a random seed of 1 (for replicability) and perform a nonparametric bootstrap on the test subset to empirically generate distributions for each of the models under evaluation. Then, we can determine whether one stochastically dominates the other, or directly estimate the probability in the frequentist manner.

6.2 Results

After 10,000 replicates of sampling with replacement for each model, I obtained RMSE estimates which are saved in plain-text files `RMSE2.txt` (for the first, parsimonious model with d=2) and `RMSE31.txt` (for the second, complicated model with d=31). As we can see in the RMSE histograms colored light blue (for d=2)

and red (for d=31), the parsimonious model stochastically dominates (by having lower RMSE out-of-sample than) the complicated model despite the reverse being true in-sample:



The mean estimated RMSE for d=2 is 0.008490 and for d=31 is 0.008718. Although both are worse out-of-sample than in-sample, the overfitting is more severe for the highly flexible, complicated model.

7 Discussion

7.1 Conclusion

The daily change in 10-year yield rate can be well-approximated by a linear fit of the corresponding changes in yield rates for bonds with similar maturity (7 and 20 years). Significantly positive correlation coefficients suggests that the response is a weighted average of the aforementioned predictors. Classical linear regression works fine for this application, and the inclusion of features based on second-order interaction terms does not lead to a meaningfully better estimator. No claims will be made regarding extrapolation beyond the present data set.

7.2 Future Work

The features in this study were limited to first and second-order polynomials of changes in yield rates on US Treasuries. There are many economic and financial indicators worthy of examination, from global commodities indices to microfinance default rates. However, from past experience, I've witnessed that the most correlated instruments are usually from within the same asset class and country of issue. Cleaning time series (especially tick data) from multiple sources can be very time-consuming, and crunching big data is slow. With sufficient computational power, we could data mine for more interaction effects, for instance, in the framework of the generalized additive model or via feature importance from the random forests algorithm for regression.